Control Systems (EEE/ECE/INSTR F242)

Tutorial 1 (13 January 2026)

1. Find the transfer function, $G(s) = I_2(s)/V(s)$, for the circuit given in Figure Q1, using nodal analysis.

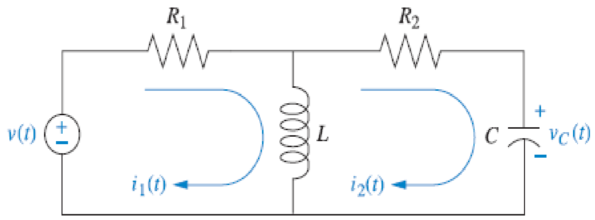

Fig. Q1

2. Write the differential equation governing the dynamic behaviour of rotational mechanical system shown in Fig. Q2. Also, find the transfer function, $G(s) = \theta_2(s)/T(s)$.

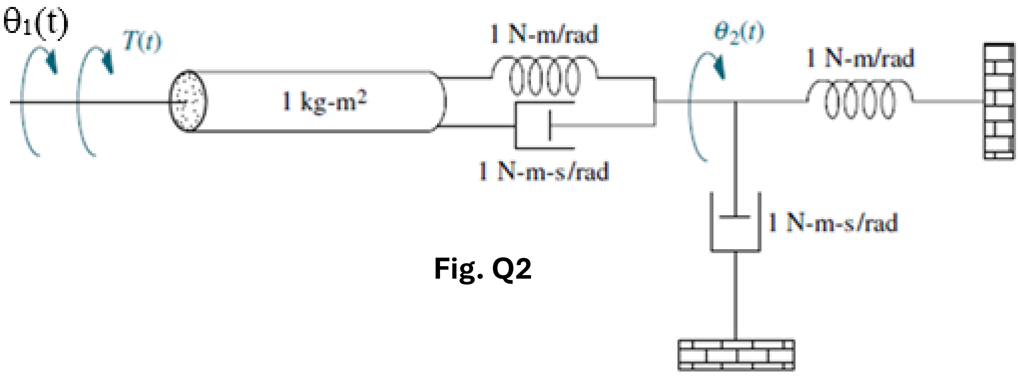

Fig. Q2

3. Write the differential equation governing the dynamic behaviour of translational mechanical system shown in Fig. Q3. Also, find the transfer function, $G(s) = X_2(s)/F(s)$.

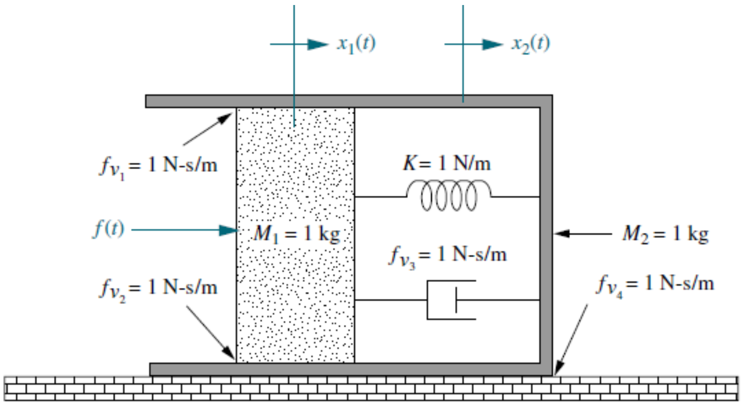

Fig. Q3

4. A rectangular bar is supported at the two ends by two springs and an external force $F(t)$ is applied to a point offset from the CG of the bar as shown in Fig. Q4. Weight of the bar is negligible compared to the force $F(t)$ and initially the bar was in a horizontal position. Determine the transfer function $\theta(s)/F(s)$, where θ is the rotation of the bar about its CG taken positive in the CW direction. Assume the mass of the bar to be m and the MI about its CG to be I.

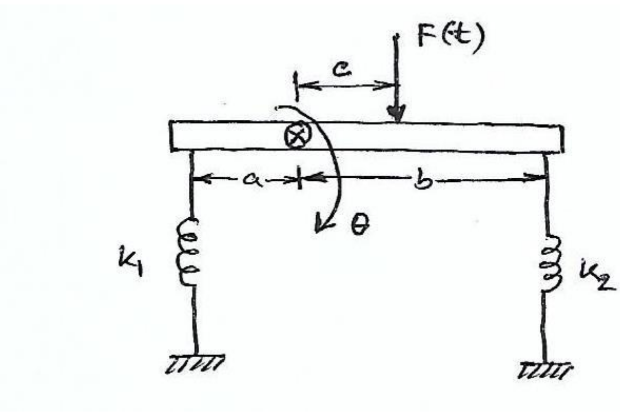

Fig. Q4

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